COR-STAT1202 Introductory Statistics
Tutorial 5
Continuous Probability Distributions and Sampling Distribution

Q1. The random variable X has probability density function as follows:

$$f(x) = \begin{cases} x & for\ 0 < x < 1 \\ 2 - x & for\ 1 < x < 2 \\ 0 & for\ all\ other\ values\ of\ x \end{cases}$$

a) Sketch the probability density function for x.
b) Show that the function has the properties of a probability density function.
c) Find the probability that x takes a value between 0.5 and 1.5.

Q2. A professor sees students during regular office hours. Time spent with students follows an exponential distribution with a mean of 10 minutes.

a) Find the probability that a given student spends fewer than 20 minutes with the professor.
b) Find the probability that a given student spends more than 5 minutes with the professor.
c) Find the probability that a given student spends between 10 to 15 minutes with the professor.

Q3. A company services home air conditioner. It is known that times for service calls follow a normal distribution with a mean of 60 minutes and a standard deviation of 10 minutes.

a) What is the probability that a single service call takes more than 70 minutes?
b) What is the probability that a single service call takes between 50 and 70 minutes?
c) The probability is 0.025 that a single service call takes more than x minutes. Determine x.
d) Find that shortest range of times that includes 50% of all service calls.
e) A random sample of four service calls is taken. What is the probability that exactly two of them take more than 65 minutes?

Q4. In one year, earnings growth of the 500 largest U.S. corporations averaged 9.2%; the standard deviation was 3.5%. Using the Empirical rule, determine the interval where it can be estimated that

a) approximately 68% of these earnings growth figures will be within this interval.
b) approximately 95% of these earnings growth figures will be within this interval.

Q5. TATA motors, Ltd. purchases computer process chips from two suppliers, and the company is concerned about the percentage of defective chips. A review of the records for each supplier indicates that the percentage defectives in consignments of chips follow normal distributions with the means and standard deviations given in the following table. The company is particularly anxious that the percentage of defectives in a consignment does not exceed 5% and wants to purchase from the supplier that's more likely to meet that specification. Which supplier should be chosen?

Supplier	Mean	Std Dev
A	4.4	0.4
B	4.2	0.6

Q6. Prairie flower Cereal, Inc., is a small but growing producer of hot and ready-to-eat breakfast cereals. You have been asked to test the cereal-packing process of 18-ounce (510 gram) boxes of sugar-coated wheat cereal. Two machines are used for the packaging process. Twenty samples of five boxes each from each machine are randomly sampled and weighted. The data are contained in the data file “Sugar Coated Wheat”. Assume normal distribution and that sample variance is a good estimate of the population variance.

a) Compute the overall sample mean and overall sample variance of the weight of cereal boxes.
b) Determine the probability that a single sample mean weight is below 500.
c) Determine the probability that a single sample mean weight is above 508.

Q7. A charity has found that 42% of all donors from last year will donate again this year. A random sample of 300 donors from last year was taken.

a) What is the standard error of the sample proportion who will donate again this year?
b) What is the probability that more than half of these sample members will donate again this year?
c) What is the probability that the sample proportion is between 0.40 and 0.45?

Q8. Assume that the study hours of students at SMU is approximately normally distributed with a mean of 20 and a standard deviation of 6.

a) Find the probability that a randomly chosen student studies less than 10 hours.
b) What is the percentage of students that study more than 35 hours.
c) A certain seminar group consists of 50 students. Assume that it forms a simple random sample from the students at SMU. Find the probability that the average number of study hours for this group is between 19 and 22 hours.

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Extra Questions

1. A dance studio conducts open classes for dance enthusiasts. The number of dancers registering for the open classes each month is normally distributed with a mean of 100 and a standard deviation of 8.
   a) What is the probability that the number of registrations in a particular month is greater than 110?
   b) The dance studio breaks even the cost for such classes when there are 98 dancers register for classes per month, what percentage of open classes would you expect the dance studio to make a profit?

2. From the past records, the university bursary fund donated by newly graduated alumni follows a normal distribution with a mean of $100 and standard deviation of $20. The manager in charge of the bursary fund intend to appeal to the alumni this year again. He believes that the donations from the alumni are independent of each other.
   a) Find the probability that the first donation it receives will be greater than $70.
   b) Find the probability that the first donation it receives will be between $85 and $125.
   c) Find the donation amount, $x, such that 5% of the donations are more than $x.
   d) Find the probability that the first donation is at least $10 more than the second.

3. Given that the height of a student population has a standard normal distribution. Discuss the height of a student in this population with regards to the average population height if the z score is:

   a) 6.5
   b) 2.2
   c) -0.14

4. The donation received by a charity organization is expected to be normally distributed with a mean of $50 and standard deviation of $6. Assume donations are independent of each other. Calculate the following:
   a) the probability that the first donation it receives will be greater than $40.
   b) the donation amount such that 5% of donations receive are more than this amount.
   c) the probability that the first donation is at least $3 more than the second.

5. A study on how much time primary six students spent studying for primary school leaving examination (PSLE) per week during the circuit breaker period follow a normal distribution with standard deviation 9 hours. A random sample of 5 students was taken in order to estimate the mean study time for the population of all primary six students.
   a) What is the probability that the sample mean exceeds the population mean by more than 2.1 hours?
   b) What is the probability that the sample mean is more than 3.2 hours below the population mean?
   c) What is the probability that the sample mean differs from the population mean by ± 4 hours.